

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document presents the functional features implemented in the OptiCrop: Smart Agricultural Production Optimization Engine. The system includes agricultural data analysis, preprocessing, machine learning-based crop prediction, and a web-based recommendation platform. Each feature is described based on its purpose, implementation status, and contribution toward providing intelligent crop recommendations and supporting sustainable farming decisions.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Collects soil and environmental parameters from users.	FindYourCrop Page	Done	High
2	User Input Form	Predicts the most suitable crop using ML algorithms.	ML Prediction Module	Done	High
3	Dataset Analysis	Performs univariate, bivariate and multivariate analysis.	Data Analysis Module	Done	Medium
4	Data Preprocessing	Handles null values, outliers and feature preparation.	Preprocessing Module	Done	High
5	K-Means Clustering	Groups similar agricultural patterns.	Clustering Module	Done	Medium
6	Logistic Regression Model	Trains model for crop recommendation prediction.	ML Model Module	Done	High
7	Flask Web Application	Integrates frontend and backend for prediction system.	Flask Backend	Done	High
8	Report Generation	Displays crop prediction results and recommendations.	Output Module	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	8
Additional / Nice-to-Have Features	Dataset visualization and clustering analysis
Features Tested & Verified	User input, prediction generation, web navigation, model loading and crop recommendation

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, About Page, FindYourCrop Form
2	Backend / Logic	3	Flask Routing, Prediction Logic, Model Loading
3	Database / Storage	2	Dataset Storage, Saved Model
4	API / Integration	2	Flask Integration, Machine Learning Model Integration
5	Security / Authentication	1	Input Validation and Error Handling